

Supramodality

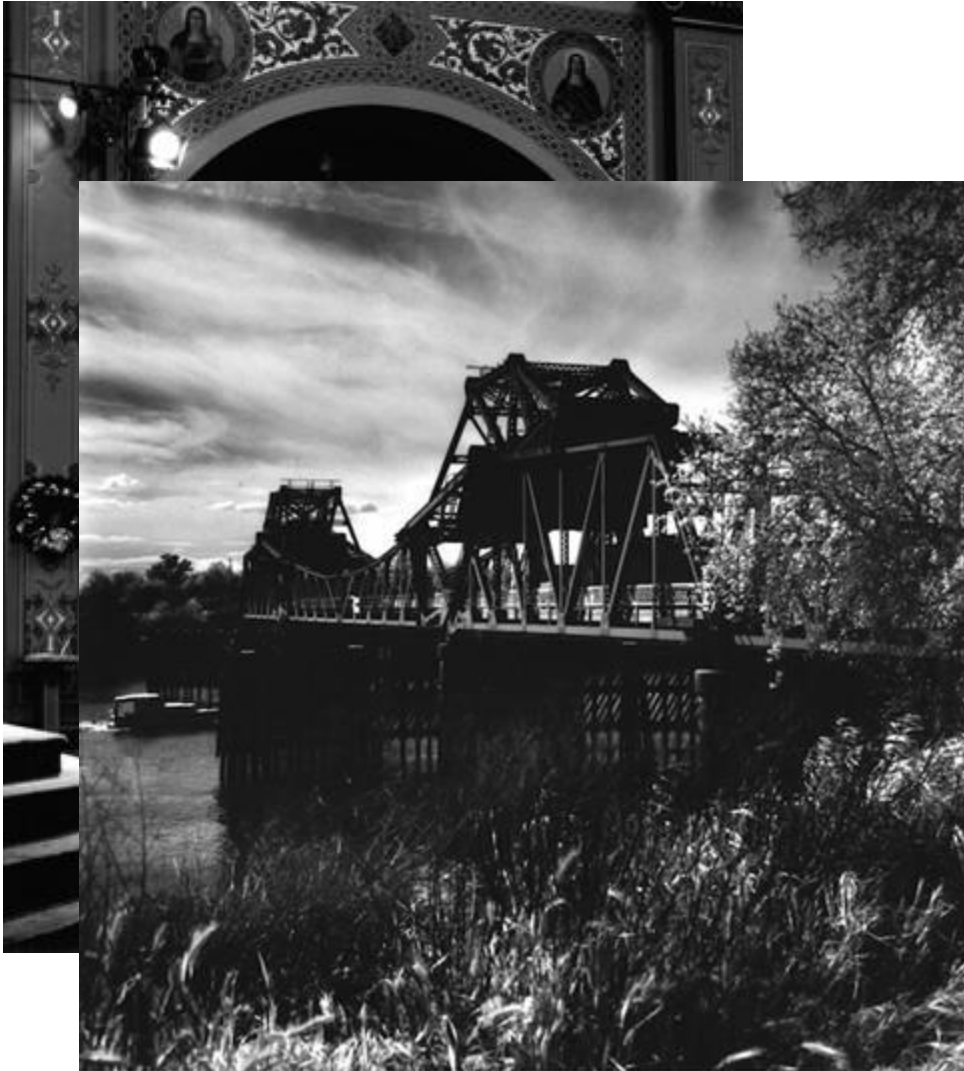


Do we really need vision?



**To what extent is vision really
necessary for the human brain
to develop and function?**

Is vision the only way "to see"?



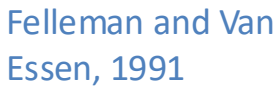
Pete Eckert

Is vision the only way "to see"?



Esref Armagan

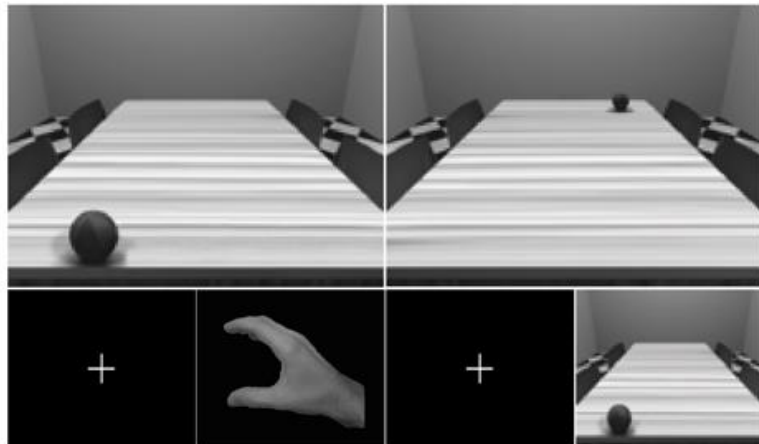
- Sight has always been regarded as the most important sense for humans to interact with the outside world



- Nonetheless, adults who are visually deprived since birth show perceptual, cognitive and social capacities that are often similar to those found in sighted individuals

Brain, vision and knowledge

Visual



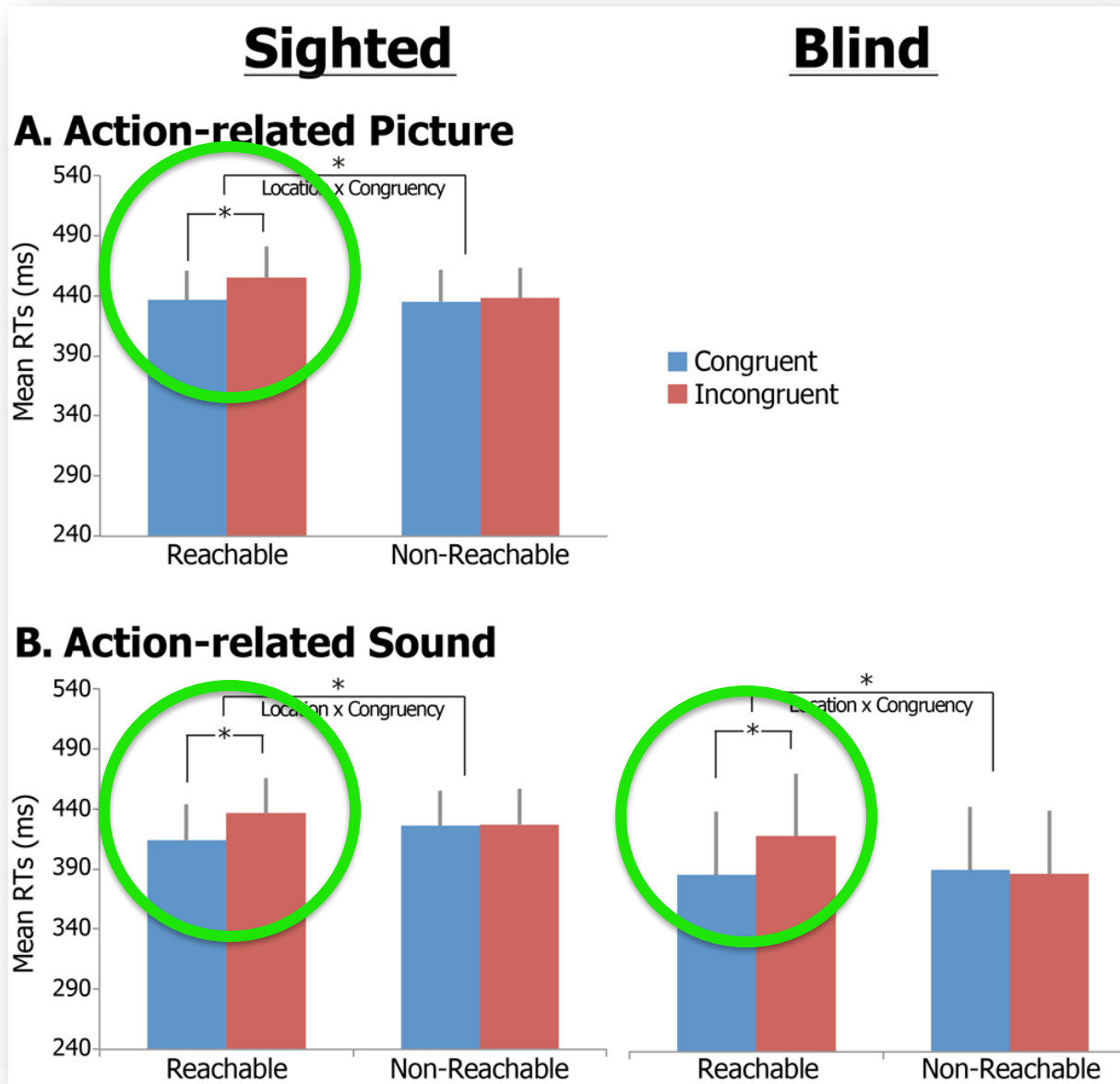
ITI (6000 ms) Instruction (150 ms) ISI (150-450 ms) Go (500 ms)

Auditory



ITI (6000 ms) Instruction (150 ms) ISI (150-450 ms) Go (800 ms)

Brain, vision and knowledge



The blind brain as a 'tool' to understand the sighted one

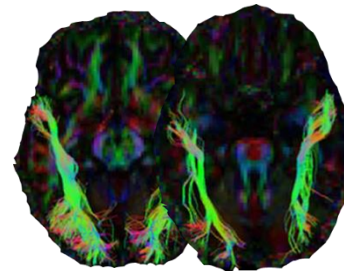
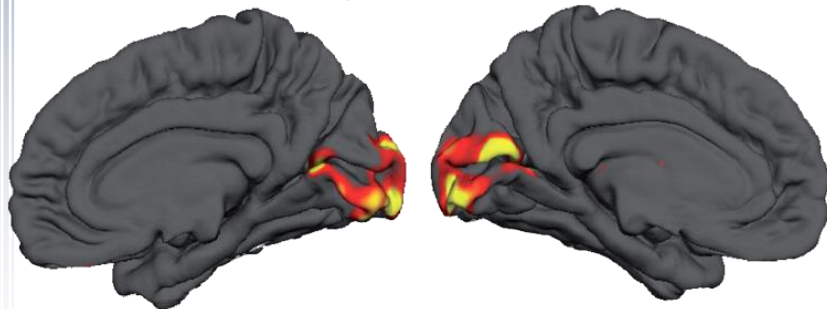


- What happens to visual-devoted brain structures in the absence of sight?
- How do individuals with congenital blindness represent a world that they have never seen?

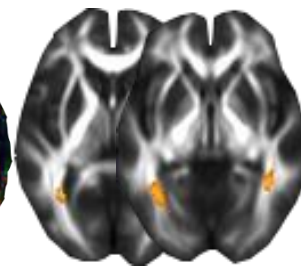
A structural reorganization

- The deafferented cortical and subcortical structures, as well as their constitutive white matter tracts undergo substantial reorganization

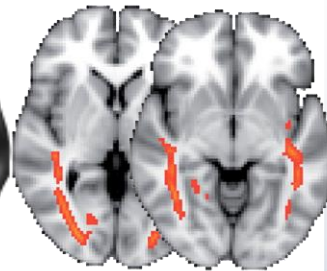
Bridge et al., 2009



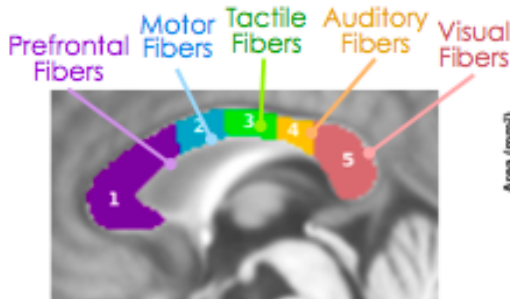
Reislev et al., 2015



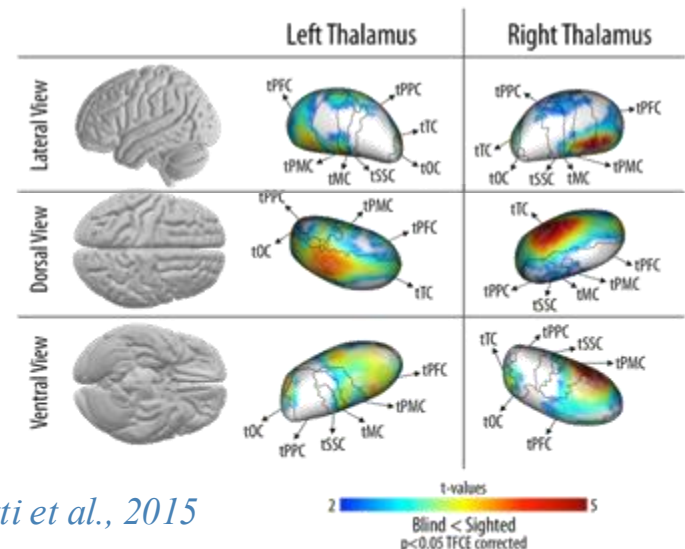
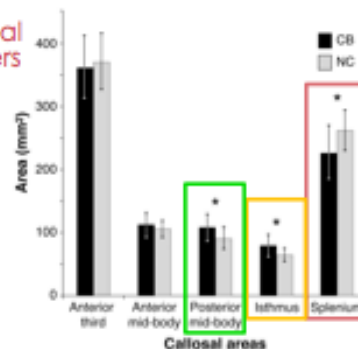
Wang et al., 2013



Bridge et al., 2009



Tomaiuolo et al., 2014



Cecchetti et al., 2015

The blind brain as a 'tool' to understand the sighted one?

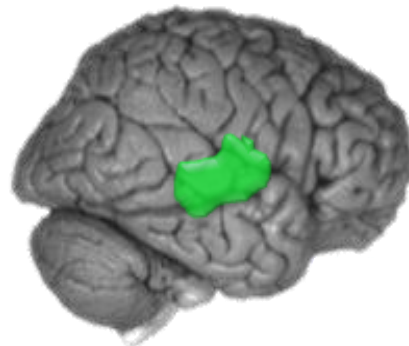
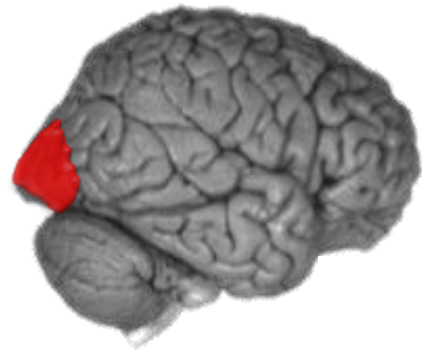
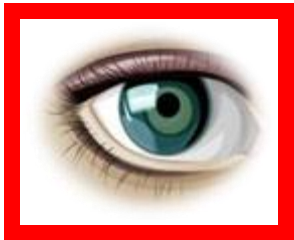


- What happens to visual-devoted brain structures in the absence of sight?
- How do individuals with congenital blindness represent a world that they have never seen?

Visual areas are used for non-visual tasks – Cross-modal (and intramodal) plasticity

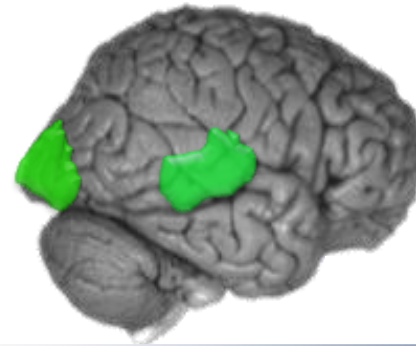
- Historically, the blind brain has been primarily investigated from the perspective of the compensatory ability of early visual areas to process non-visual information

Sighted

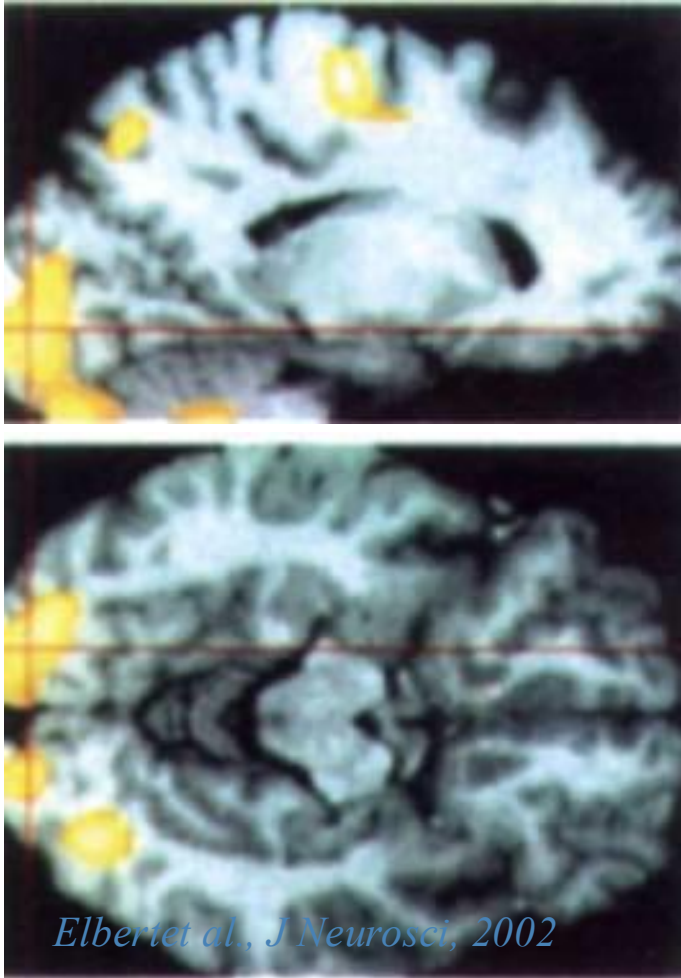


Blind

cross-modal is what happens *because of* lack of vision



Cross-modal (and intramodal) plasticity



Elbert et al., J Neurosci, 2002

Sadato et al., Nature, 1996

- Effects of task difficulty
- Correlation with specific cognitive/perceptual performances
- Real or 'virtual' lesions of occipital cortex
- Sensory overlapping
- Connectivity measures
- Functional recruitment in early but not late blind individuals

Still many open questions on cross-modality

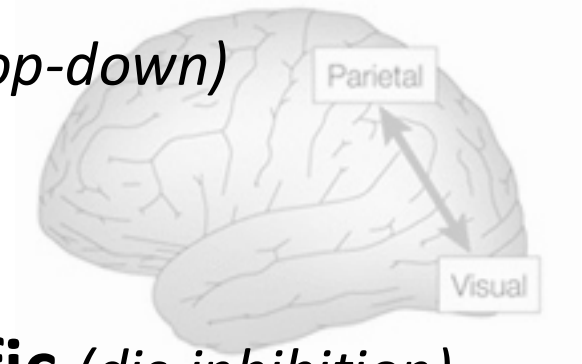
*“A growing body of evidence demonstrates that the brain can reorganize dramatically following sensory loss. Although the existence of such neuroplastic crossmodal changes is not in doubt, the **functional significance of these changes remains unclear**”*

Amy Kalia Singh et al., 2018

- *Where* in the occipital cortex?
- *How* in the occipital cortex:
unmasking vs. novel connections?
- A modulation of the non-visual modality?
- Truly compensatory?
- Which function can go cross-modal? Is cross-modality functionally-specific?

Still many open questions on cross-modality

Aspecific (*top-down*)



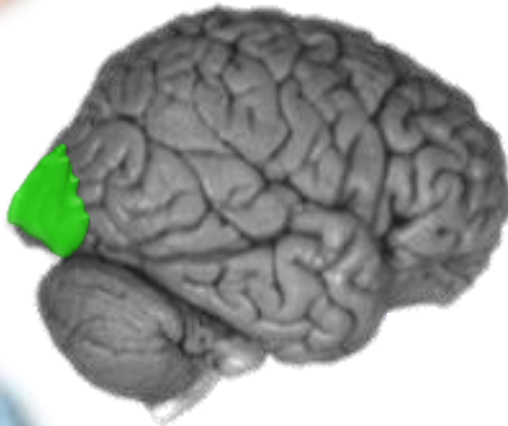
Aspecific (*dis-inhibition*)



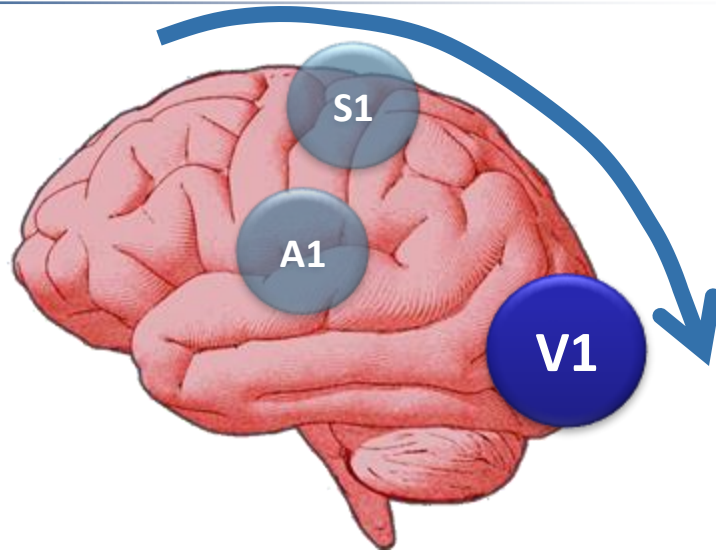
Task-specific (*compensatory*)



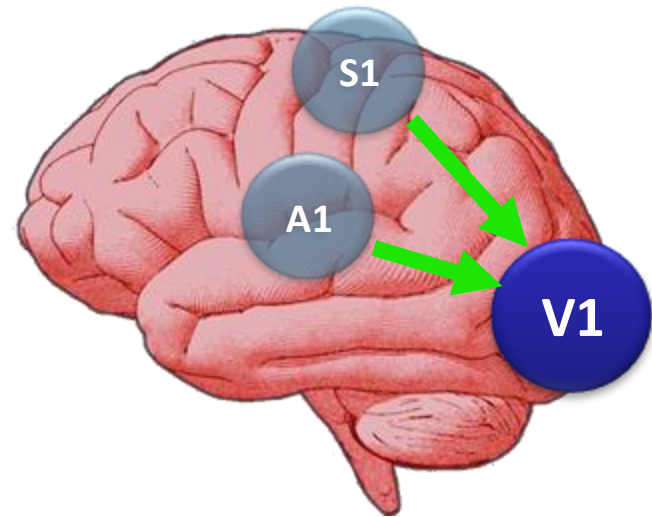
Pluripotent (*higher cognitive functions*)



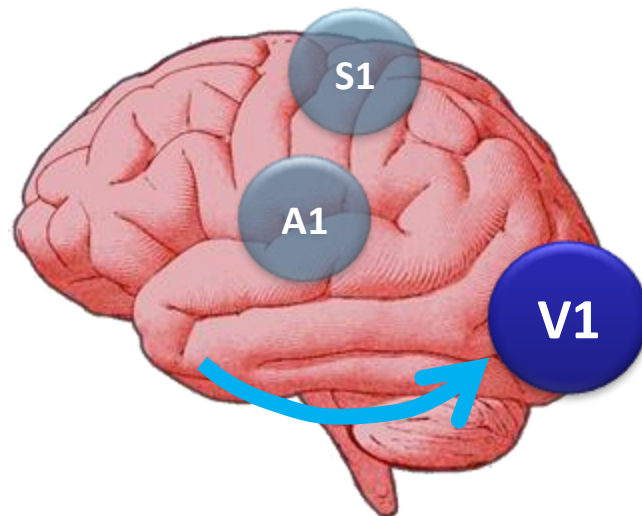
Rethinking the sensory-deprived brain?



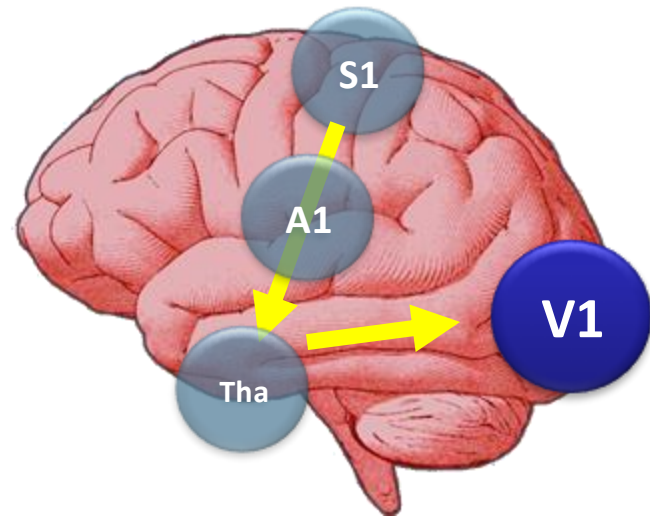
Top-down modulation



Heteromodal connections



Feedback connections



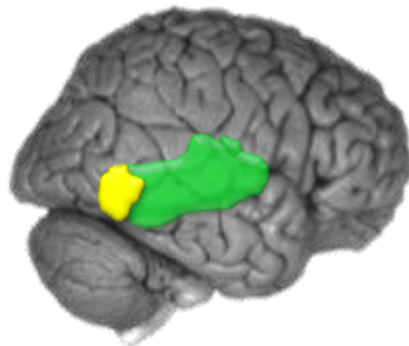
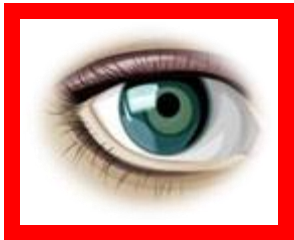
(Cortico-)thalamic-cortical connections

Supramodality (alias metamodality, polimodality, multimodality, etc.)

«If a given feature is also present in sighted individuals, its recruitment in congenitally blind individuals reflects a more abstract representation of a specific content of information, and cannot be simply a consequence of a plastic rearrangement»

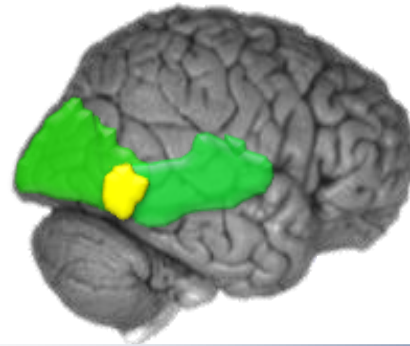
Ricciardi and Pietrini, Curr Opin Neurol, 2011

Sighted



Blind

supramodal is what
takes place *despite* the lack
of vision



Beyond sensory images: Object-based representation in the human ventral pathway

5658-5663 | PNAS | April 13, 2004 | vol. 101 | no. 15

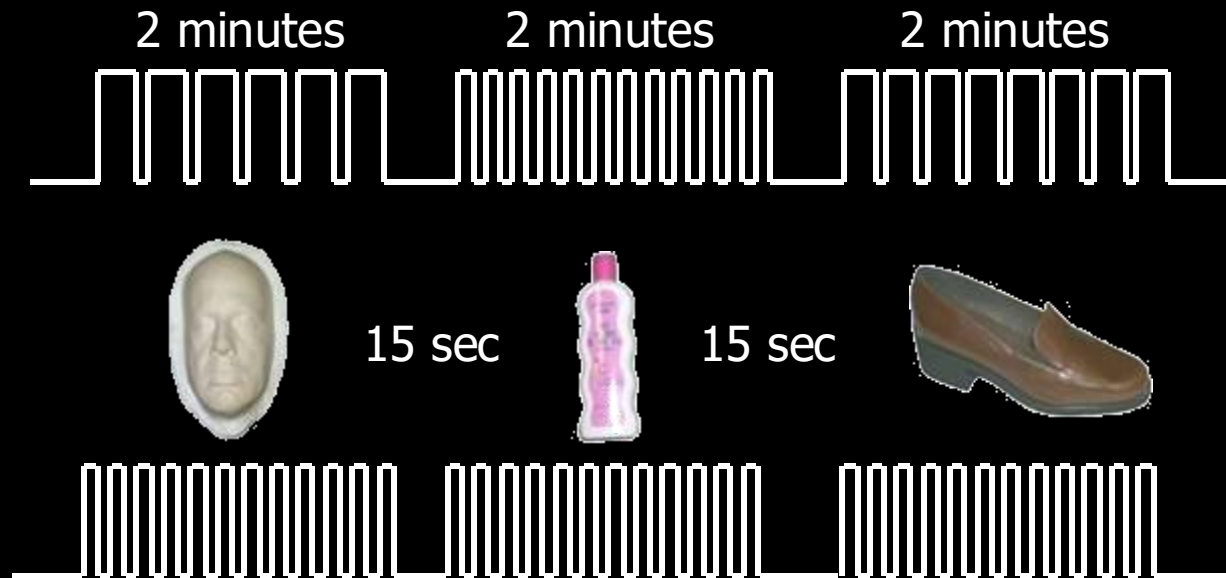
Pietro Pietrini^{*†}, Maura L. Furey^{**†}, Emiliano Ricciardi^{*}, M. Ida Gobbini^{§¶}, W.-H. Carolyn Wul[‡], Leonardo Cohen[‡], Mario Guazzelli^{**}, and James V. Haxby^{§¶}

Sighted and congenitally blind individuals

fMRI: 3 Tesla GE scanner

(GRE-EPI, TR= 2.5 sec, FOV= 24 cm, 3.5-mm sagittal slices, 8 runs of 174 volumes)

One-back



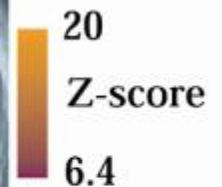
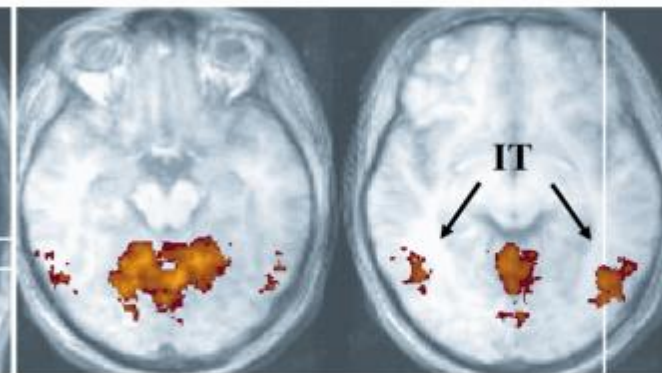
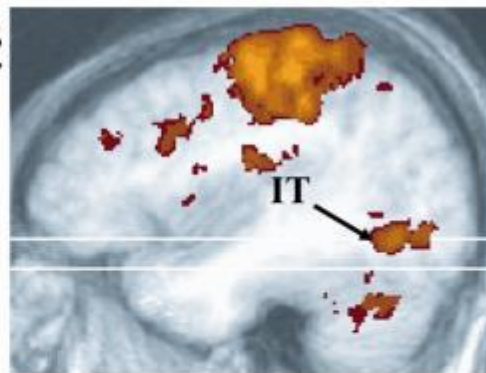
Passive perception



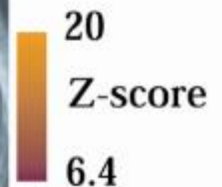
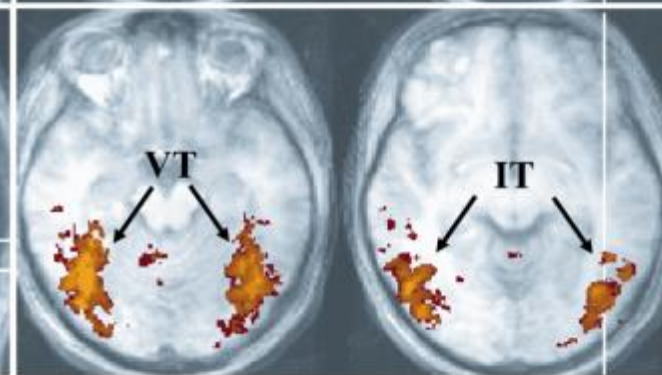
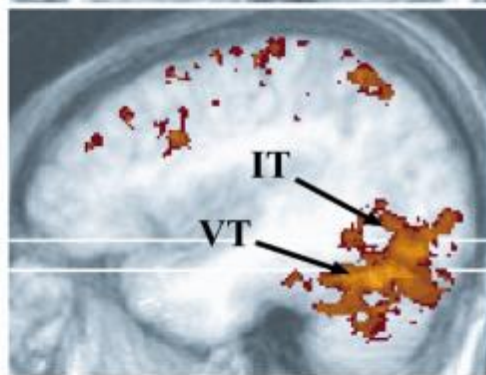
Beyond sensory images: Object-based representation in the human ventral pathway

Sighted Subjects

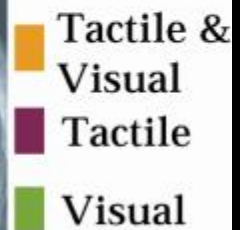
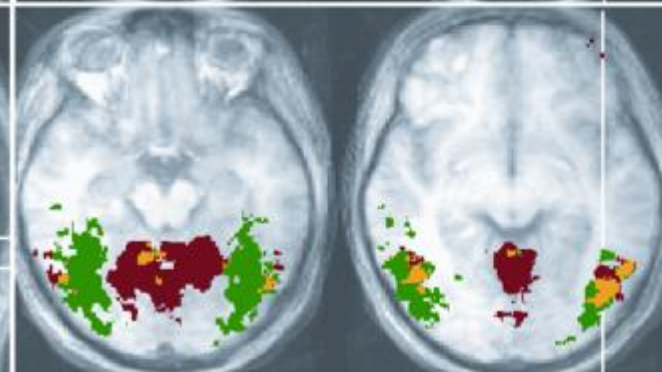
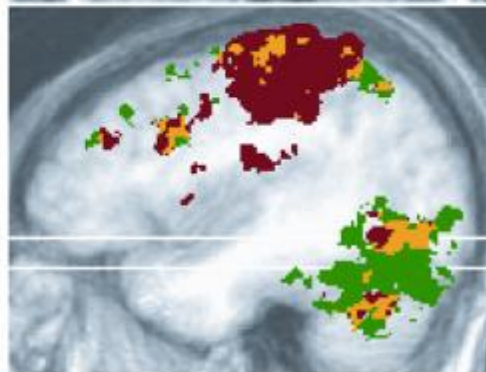
Tactile



Visual



Tactile/
Visual
Overlap



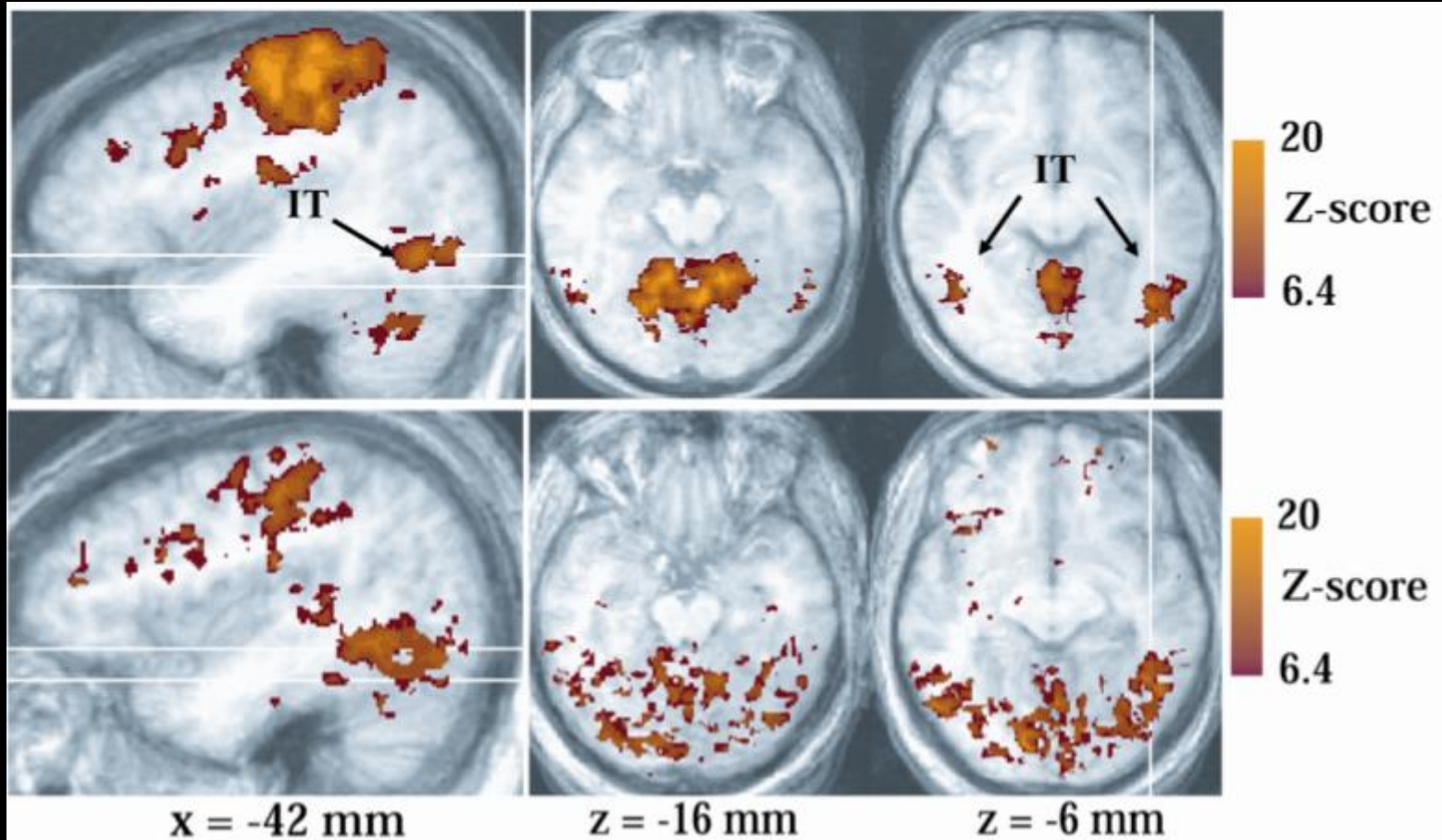
x = -42 mm

z = -16 mm

z = -6 mm

Beyond sensory images: Object-based representation in the human ventral pathway

Sighted



Blind

Object based representation in the ventral stream - Summary

- Ventral extrastriate cortex shows category-specific responses to the tactile exploration of objects during a discrimination task
- Cortical areas involved in visual discrimination of objects are also involved in tactile discrimination of the same objects
- Category-related patterns of response also in blind individuals who have had no visual experience
 - ÷ no merely due to visual-based imagery
 - ÷ visual experience is not a pre-requisite for development of category-related patterns of neural response

Optic Flow

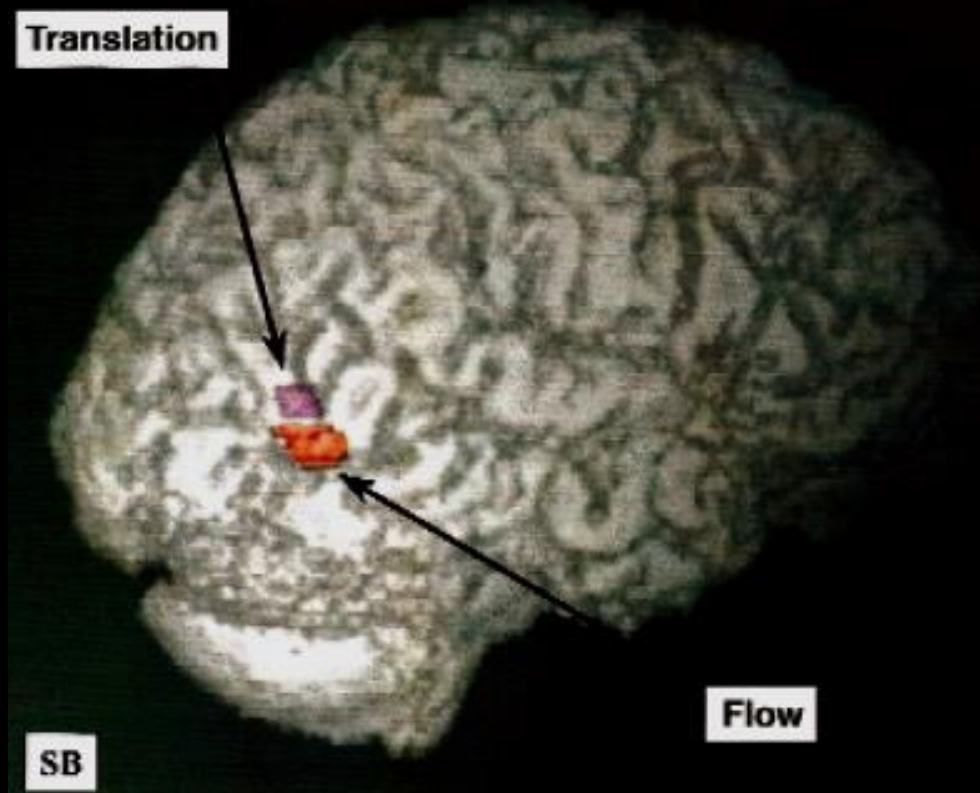


Tactile Flow

A cortical area that responds specifically to optic flow, revealed by fMRI

M. C. Morrone^{1,6}, M. Tosetti², D. Montanaro^{2,3}, A. Fiorentini¹, G. Cioni^{2,4} and D. C. Burr^{1,5}

Translation



Flow

SB

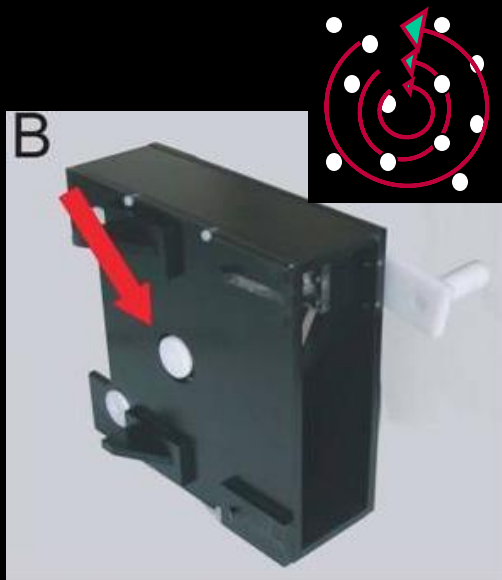
Does tactile flow also rely on hMT+ complex response?

Subjects: 6 sighted and 4 congenitally blind healthy right-handed volunteers (26 ± 3 yr)

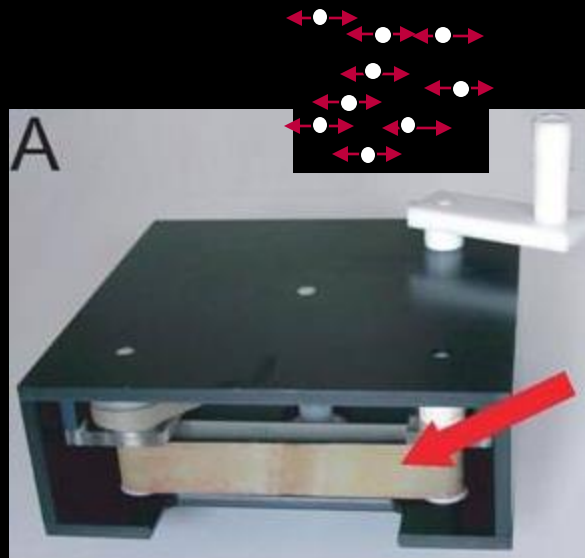
fMRI: 1.5 Tesla GE scanner GRE-EPI
TR= 3.0 sec, FOV= 24 cm 22-26 5-mm axial slices, 10-12 runs of 100 volumes

High-Res T1-weighted spoiled gradient

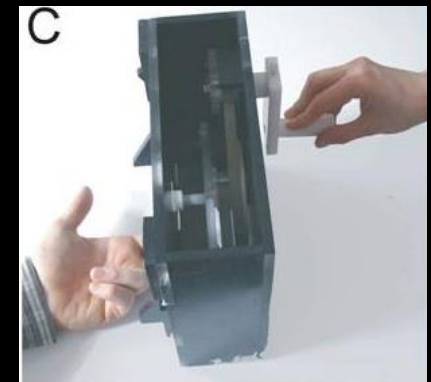
124 1.2-mm sagittal slices, FOV=24 cm



Rotational flow



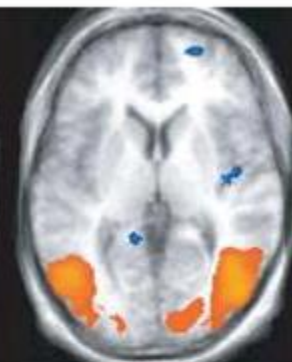
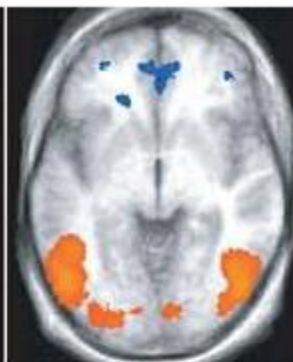
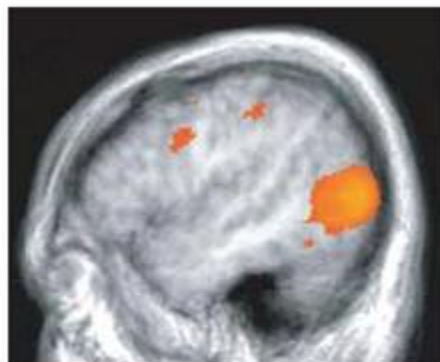
Translational flow



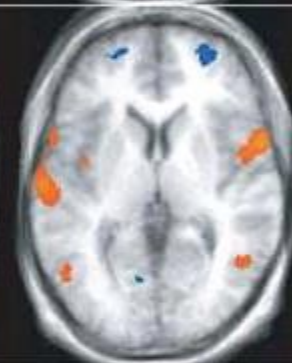
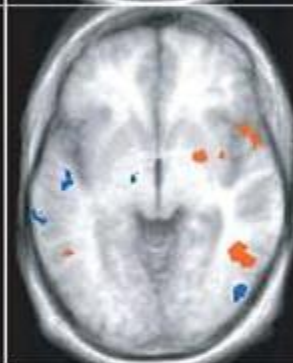
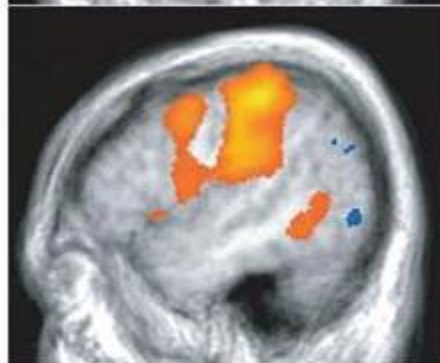
hMT+ is activated both by optic and tactile flows in sighted subjects

Sighted Subjects

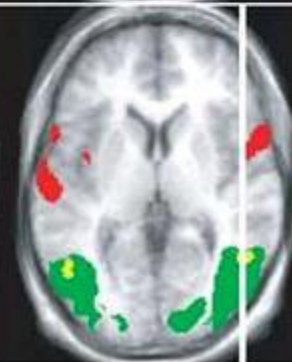
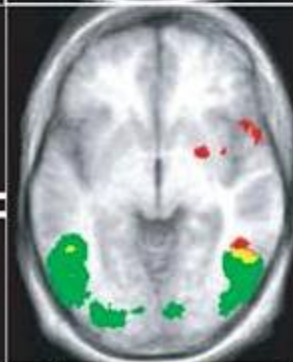
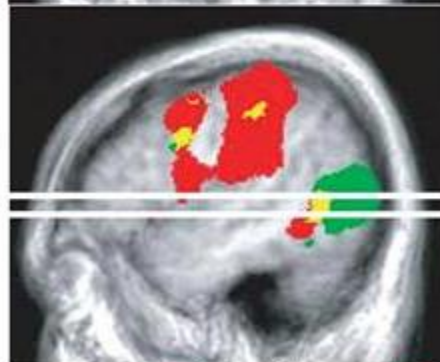
Optic Flow



Tactile Flow



Tactile/Optic
Overlap



■ Tactile

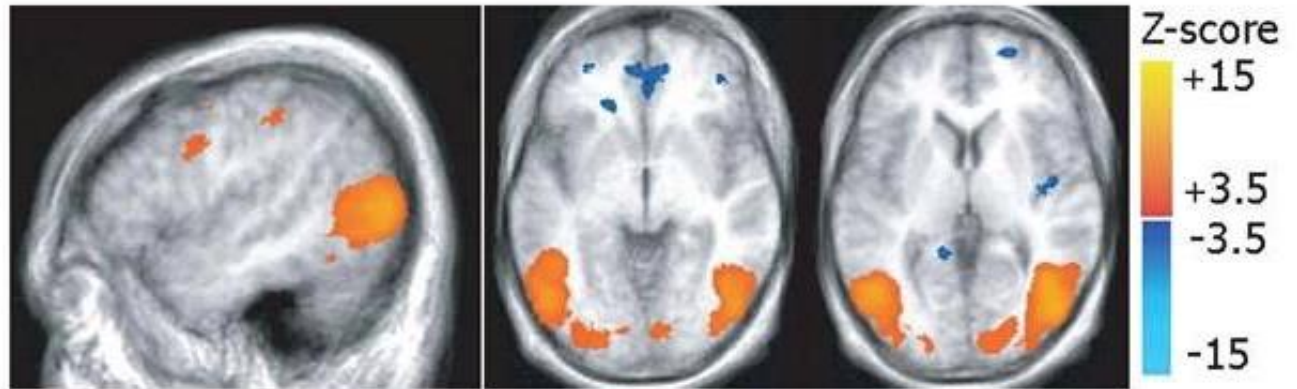
■ Optic

■ Overlap

Tactile flow activates hMT+ in congenitally blind subjects

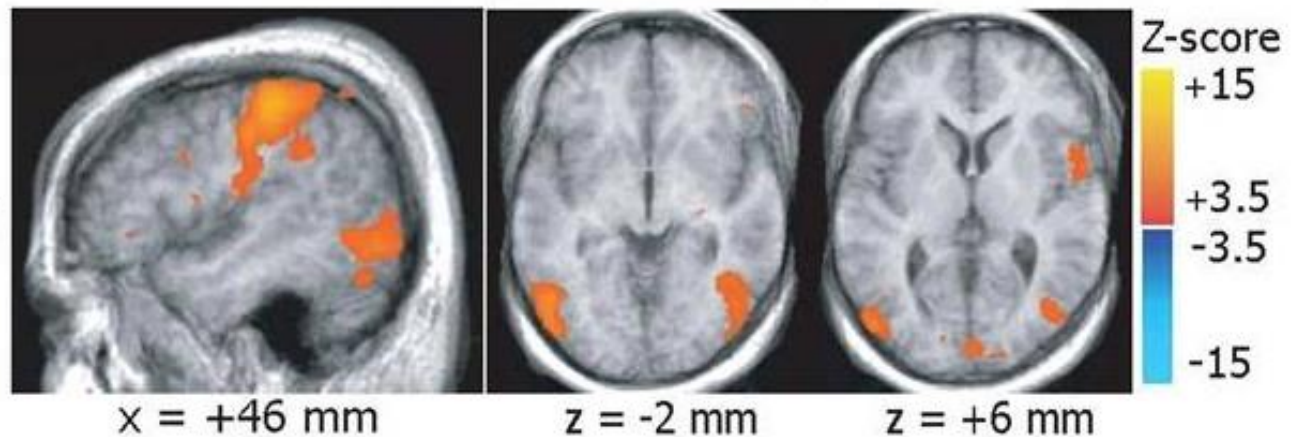
Sighted Subjects

Optic Flow



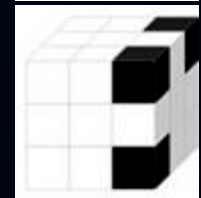
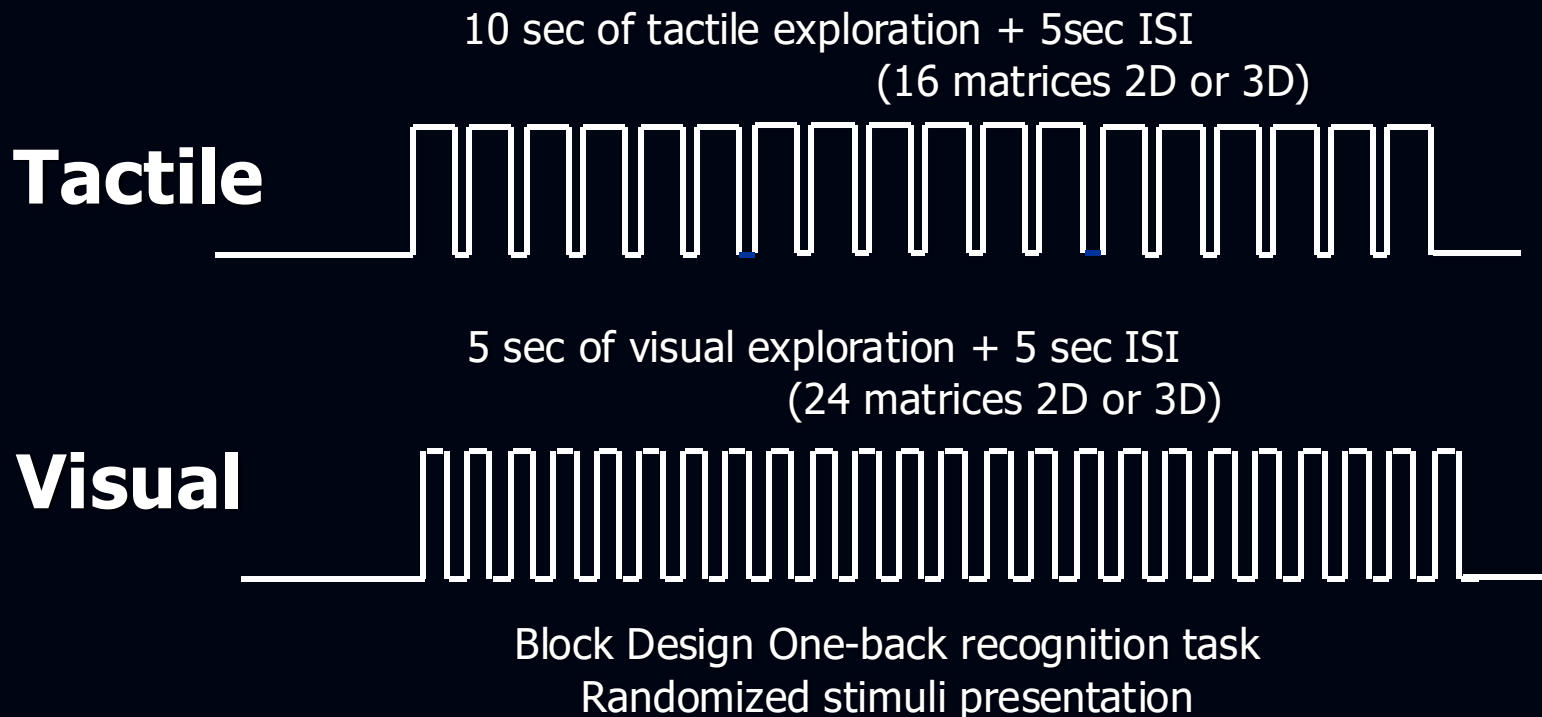
Blind Subjects

Tactile Flow



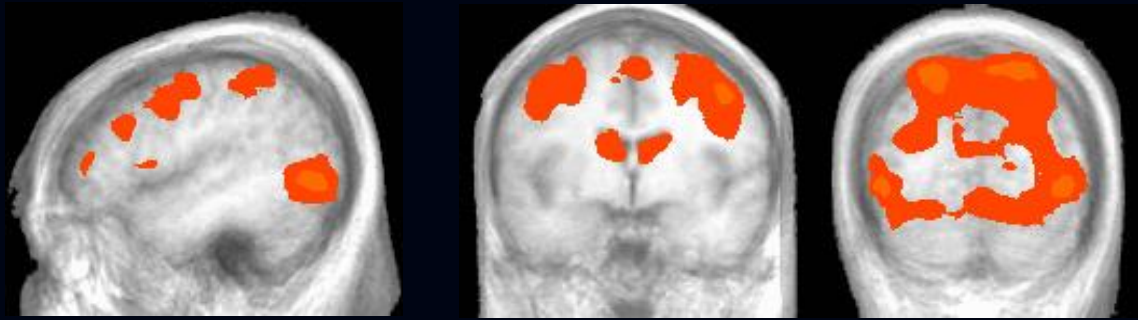
Dorsal stream response to visual and tactile spatial location tasks

- **Subjects:** 6 sighted and 4 congenitally blind right-handed healthy volunteers (mean age $\pm$ sd= 28 $\pm$ 1.3)
- **MRI:** 1.5 Tesla GE scanner Gradient Echo EPI
 - TR=2500 msec, FOV= 24 cm, 22-26 5-mm thick axial slices, TE=40 msec, FA=90° , IPR=64x64 pixels, 6-8 time series of 114 volumes
 - high-resolution T1-weighted SPGR



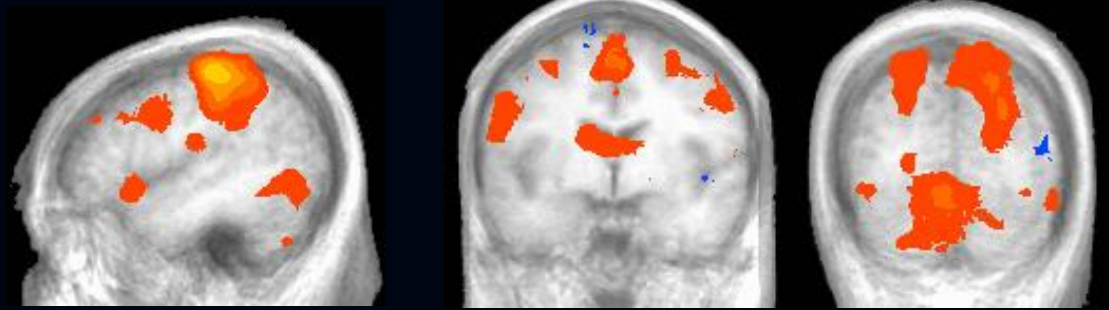
Parieto-occipital cortex is activated both by optic and tactile spatial detection in sighted

Visual



Tactile

+22 Z score -22

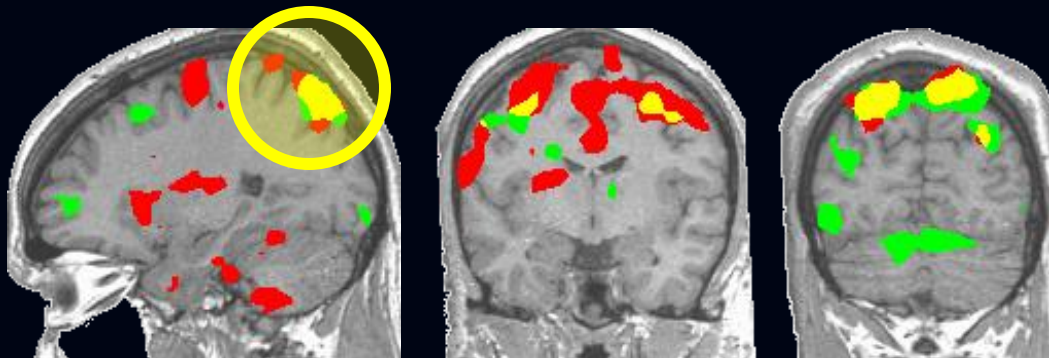


x = -46 mm

y = -2 mm

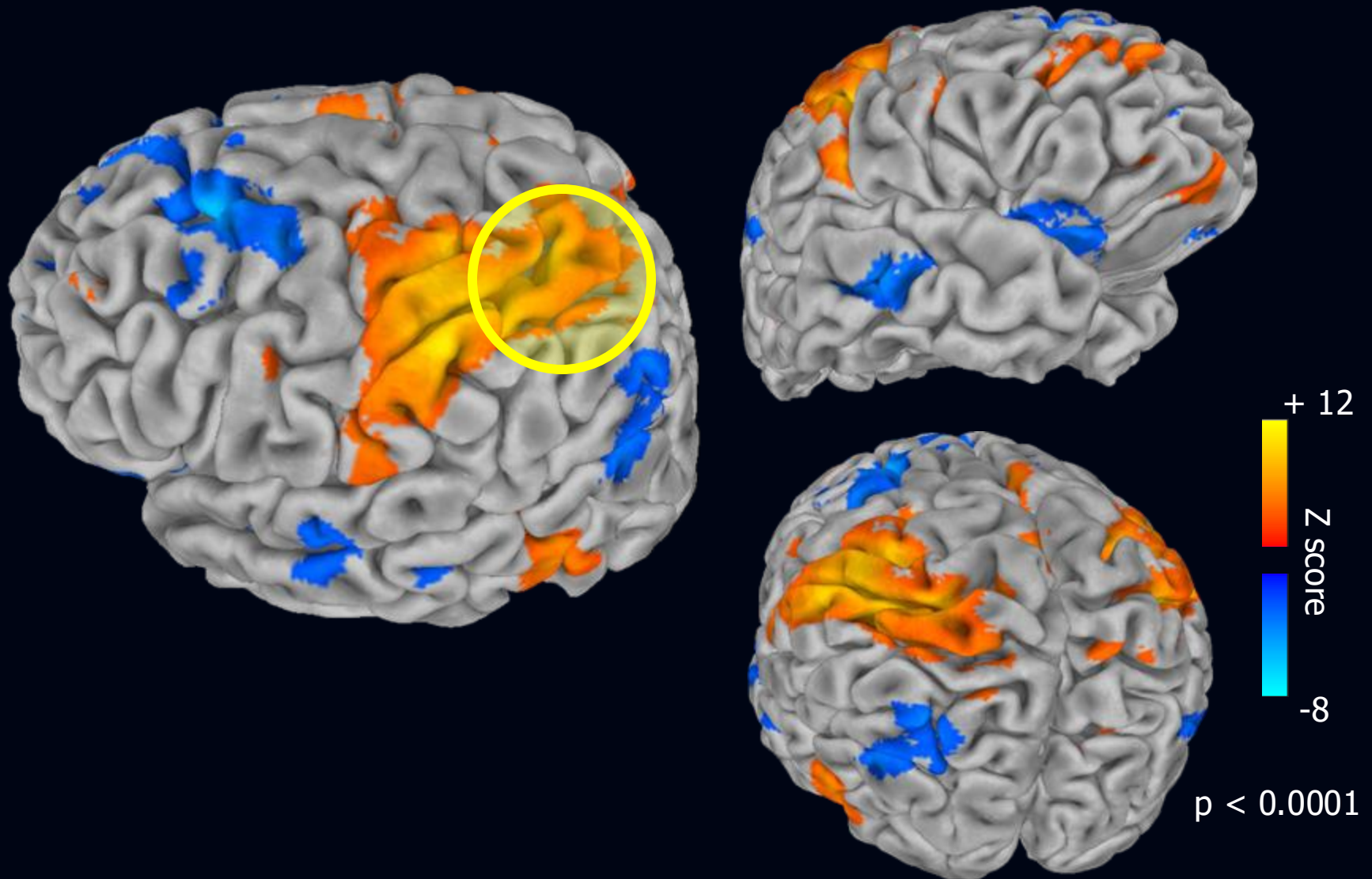
y = -72 mm

Visual
and
tactile



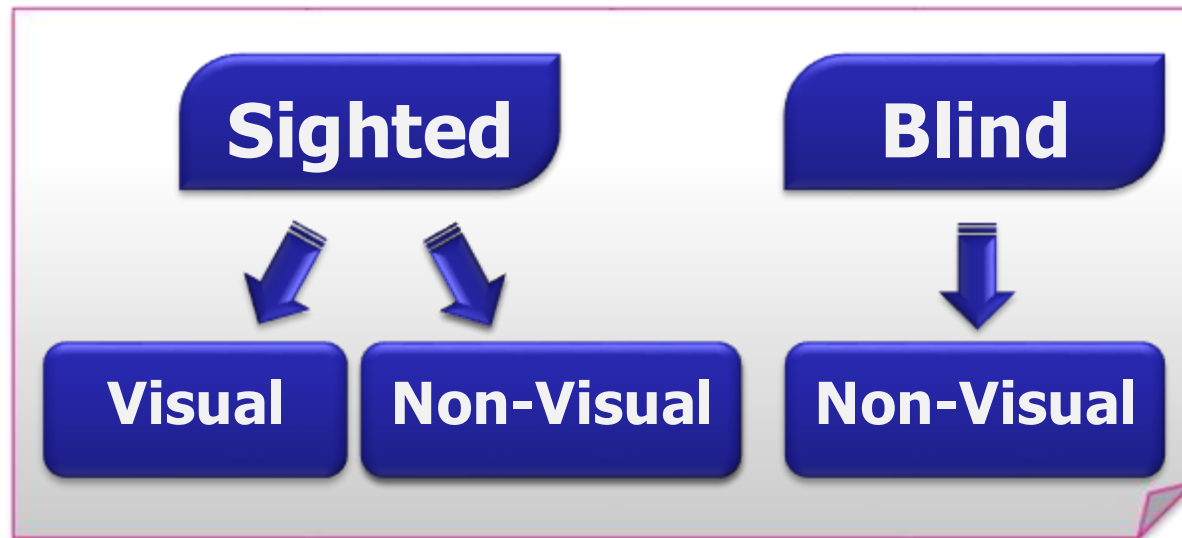
■ Tactile
■ Visual
■ Tactile ∩ Visual

Parieto-occipital cortex is activated by tactile spatial detection in blind

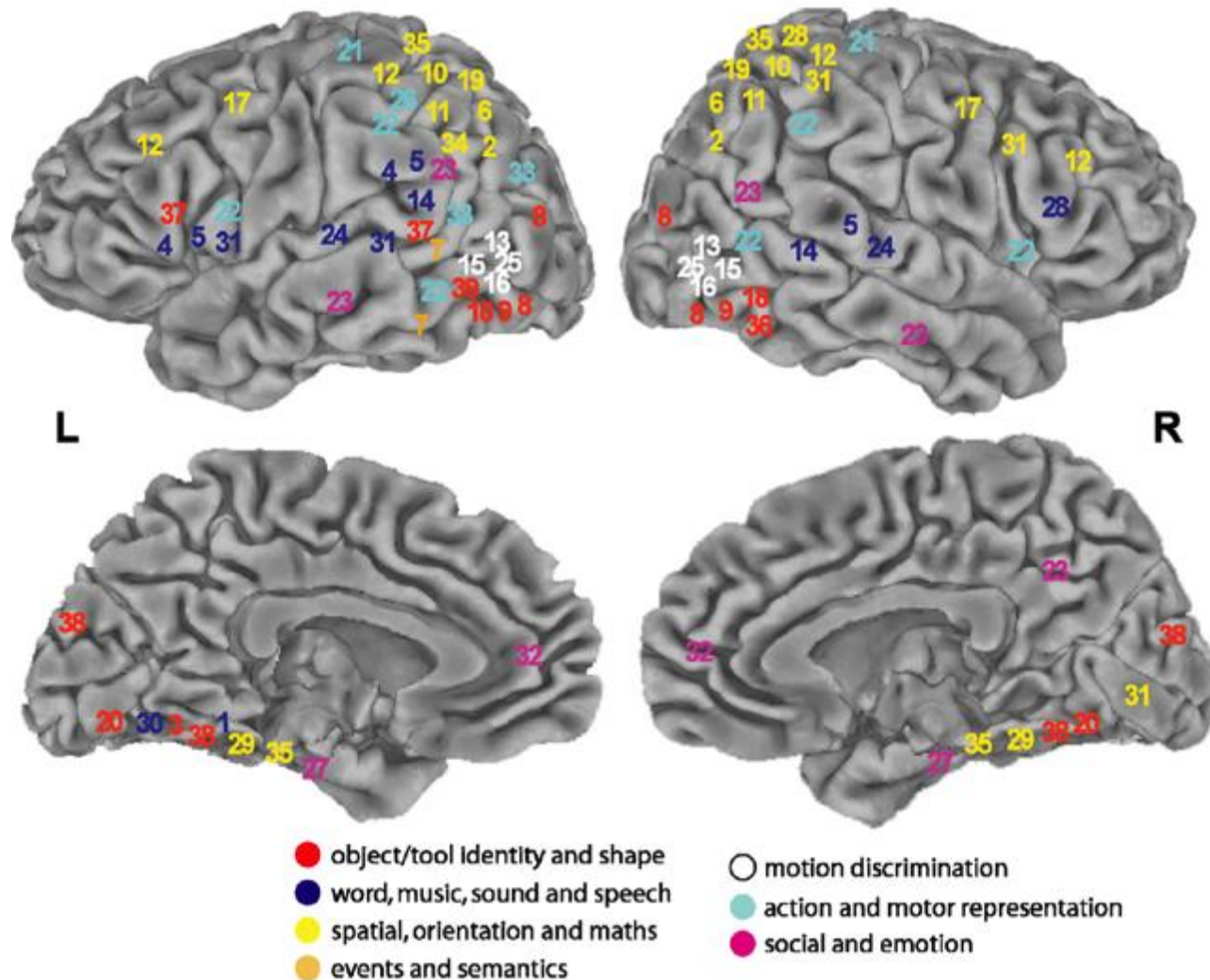


Supramodal vs. cross-modal response

«At the same time, if a given feature **is also present in sighted individuals, its functional recruitment in congenitally blind individuals has to reflect a more abstract, supramodal representation of a specific content of information**, either structurally or semantically, and cannot be simply a consequence of a plastic rearrangement due to the lack of vision»



Vision is not a mandatory prerequisite for the brain cortical organization to develop and function



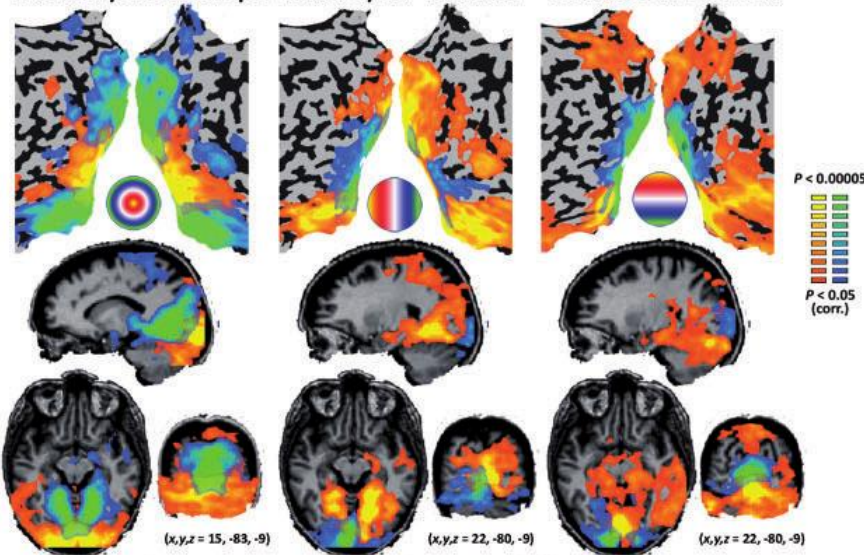
Vision is not a mandatory prerequisite for the brain cortical organization to develop and function

Functional connectivity of visual cortex in the blind follows retinotopic organization principles

Ella Striem-Amit,^{1,2,*} Smadar Ovadia-Caro,^{3,4,*} Alfonso Caramazza,^{2,5} Daniel S. Margulies,^{3,4} Arno Villringer^{3,4} and Amir Amedi^{1,6,7,8}

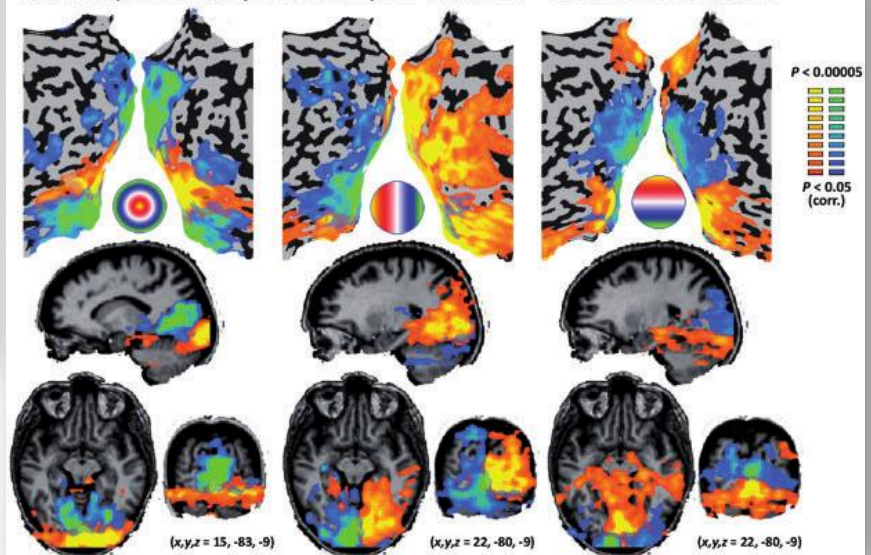
B fMRI retinotopic organization in sighted controls

Eccentricity: cVF versus pVF Laterality: rVF versus lVF Elevation: bVF versus tVF

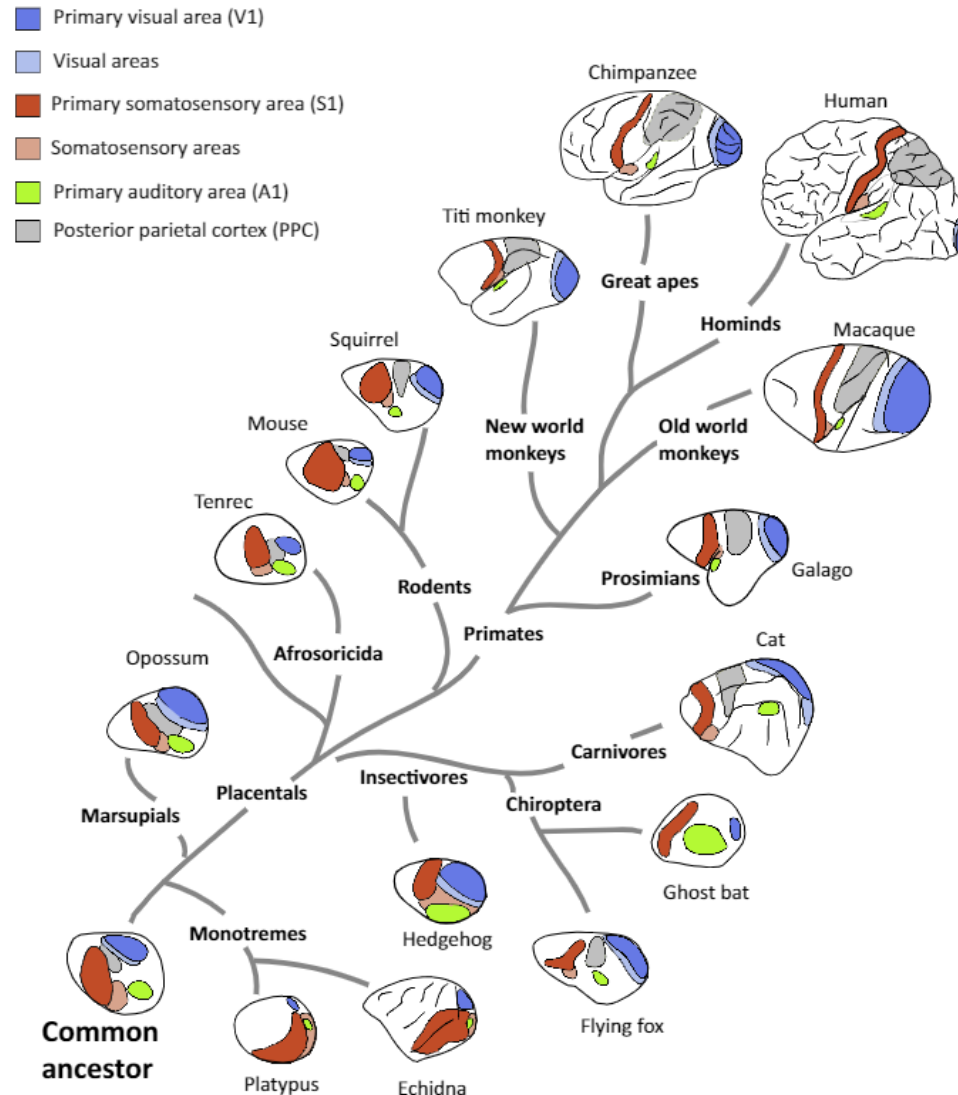


C fMRI retinotopic organization in congenitally blind people

Eccentricity: cVF versus pVF Laterality: rVF versus lVF Elevation: bVF versus tVF

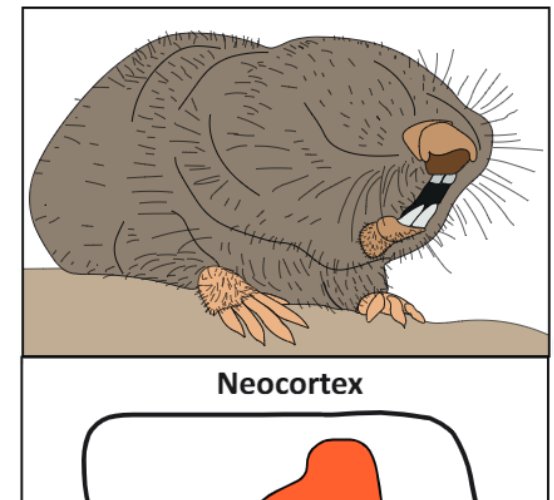


Vision is not a mandatory prerequisite for the brain cortical organization to develop and function

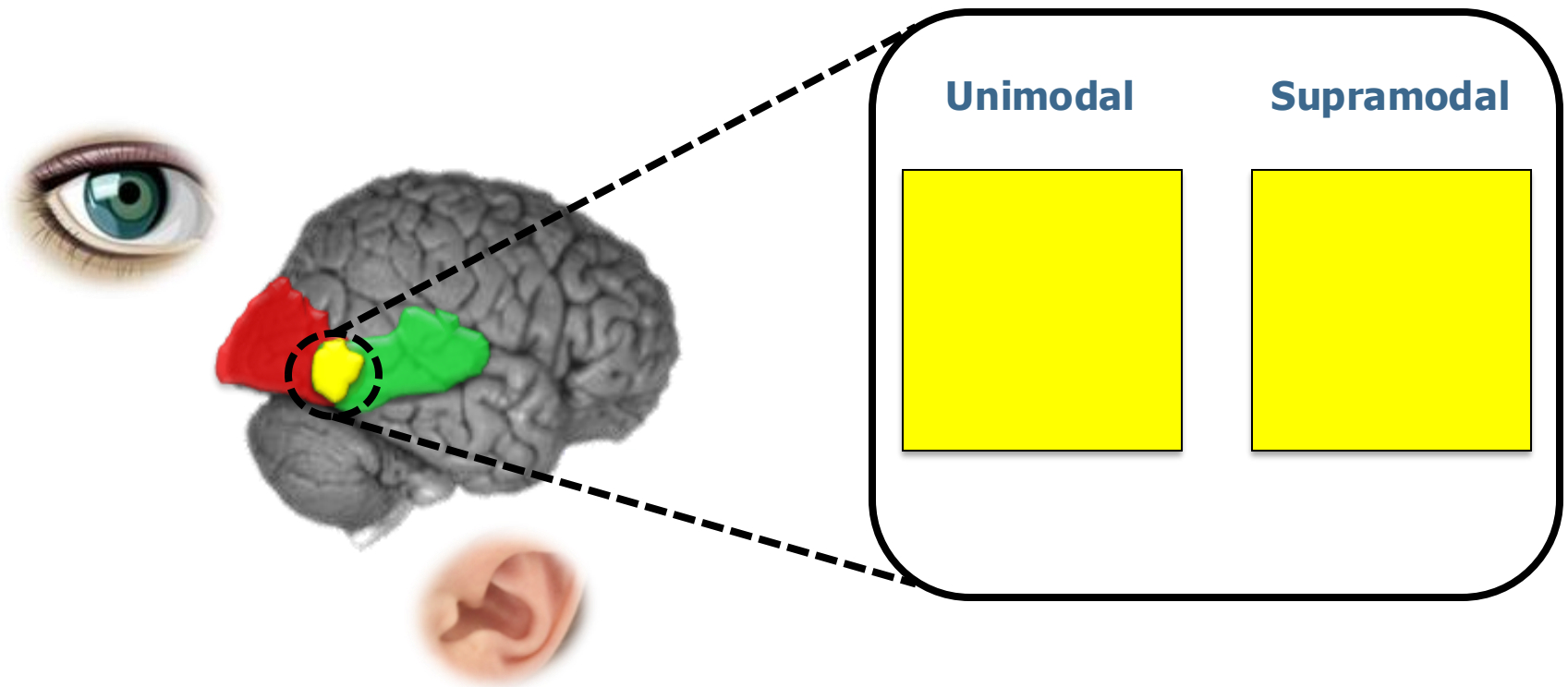


Trends in Neurosciences

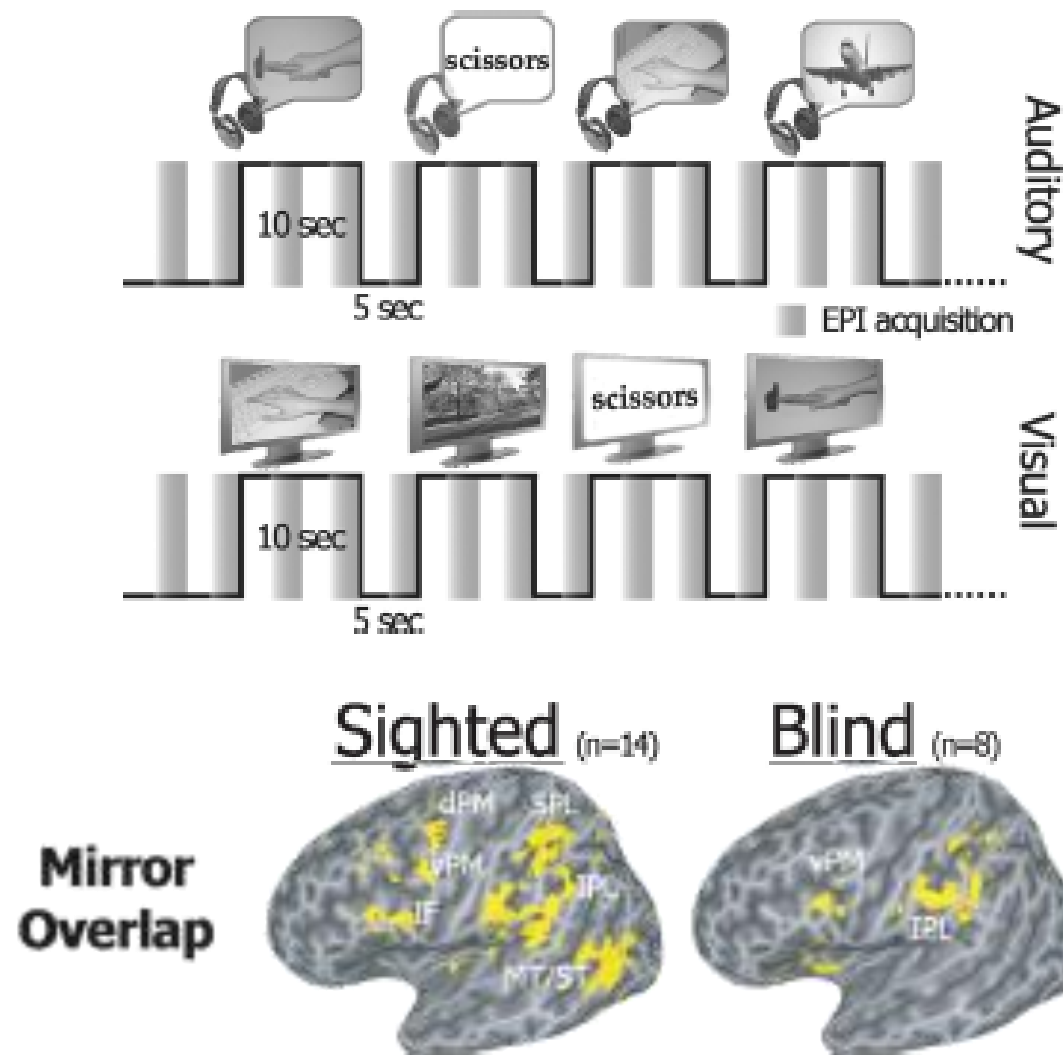
Blind mole rat



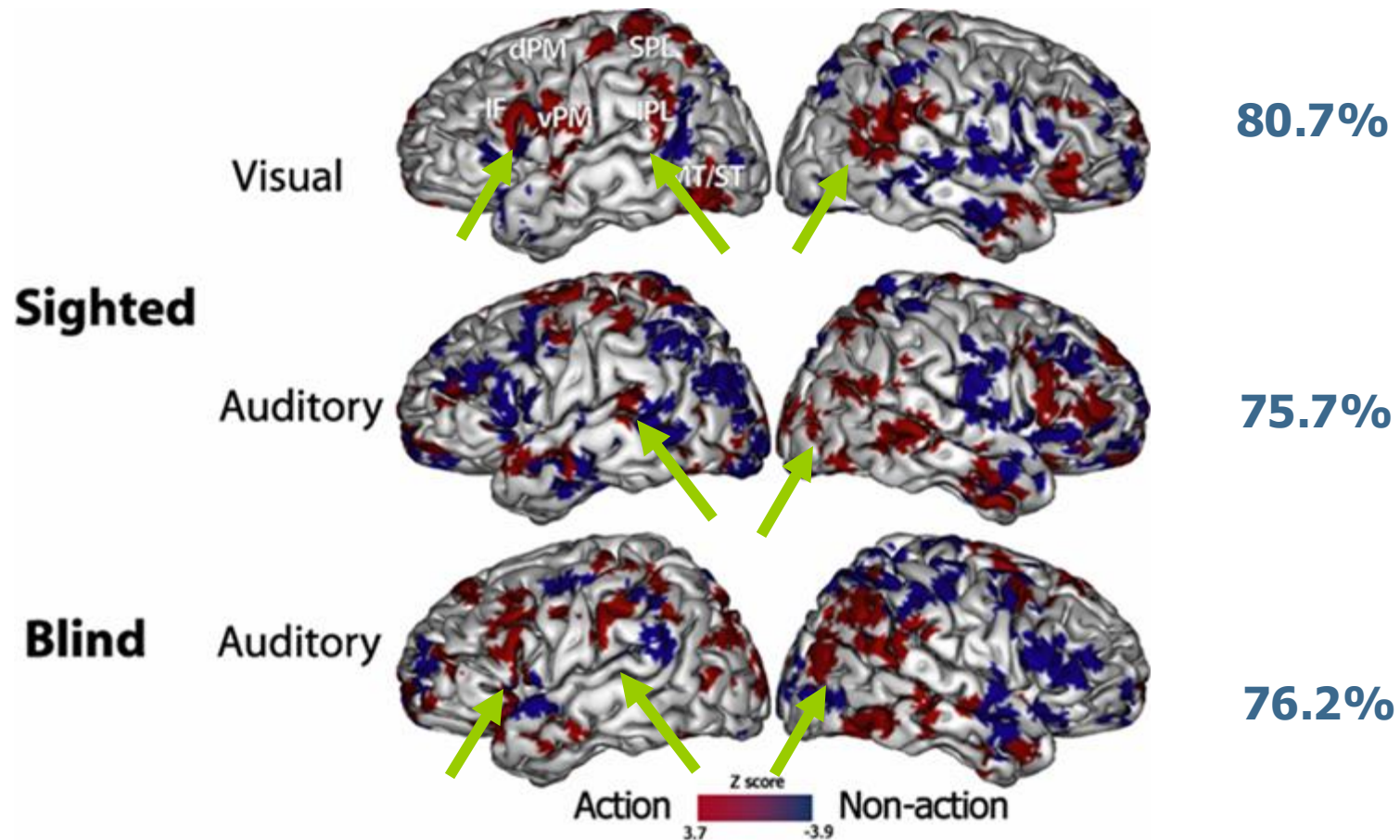
Is the more abstract representation of information truly supramodal, i.e. shares a common coding across sensory modalities?



How blind people 'see' the actions of others



Beyond motor scheme: A supramodal distributed representation in the action-observation network



A 'machine learning' classifier discriminates a supramodal and distributed functional representation of action

Beyond Motor Scheme: A Supramodal Distributed Representation in the Action-Observation Network

Emiliano Ricciardi , Giacomo Handjaras, Daniela Bonino, Tomaso Vecchi, Luciano Fadiga, Pietro Pietrini

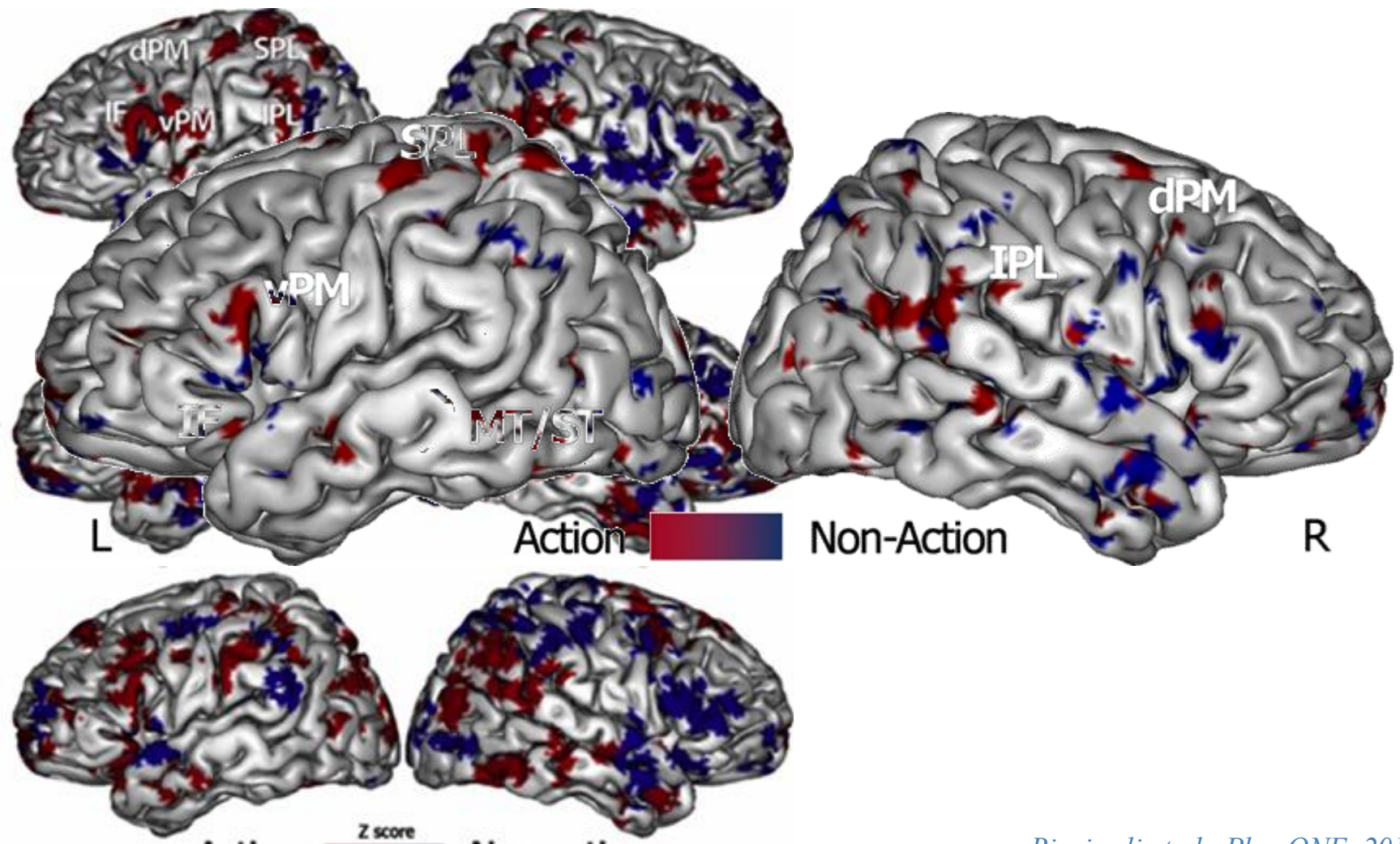
		Sighted	Blind
		SVM classifier trained on visual stimuli	SVM classifier trained on auditory stimuli
A. Whole brain			
Sighted	Visual	80.7%***	n.s.
	Auditory	n.s.	75.7%***
Blind	Auditory	n.s.	76.2%***
B. After excluding the knock-out map			
Sighted	Visual	77.1%***	n.s.
	Auditory	n.s.	74.3%***
Blind	Auditory	n.s.	73.7%***
C. By restricting to the knock-out supramodal map			
Sighted	Visual	73.6%***	61.1%**
	Auditory	59.6%*	67.1%***
Blind	Auditory	60.3%**	70.0%***

***p-value < 0.005,

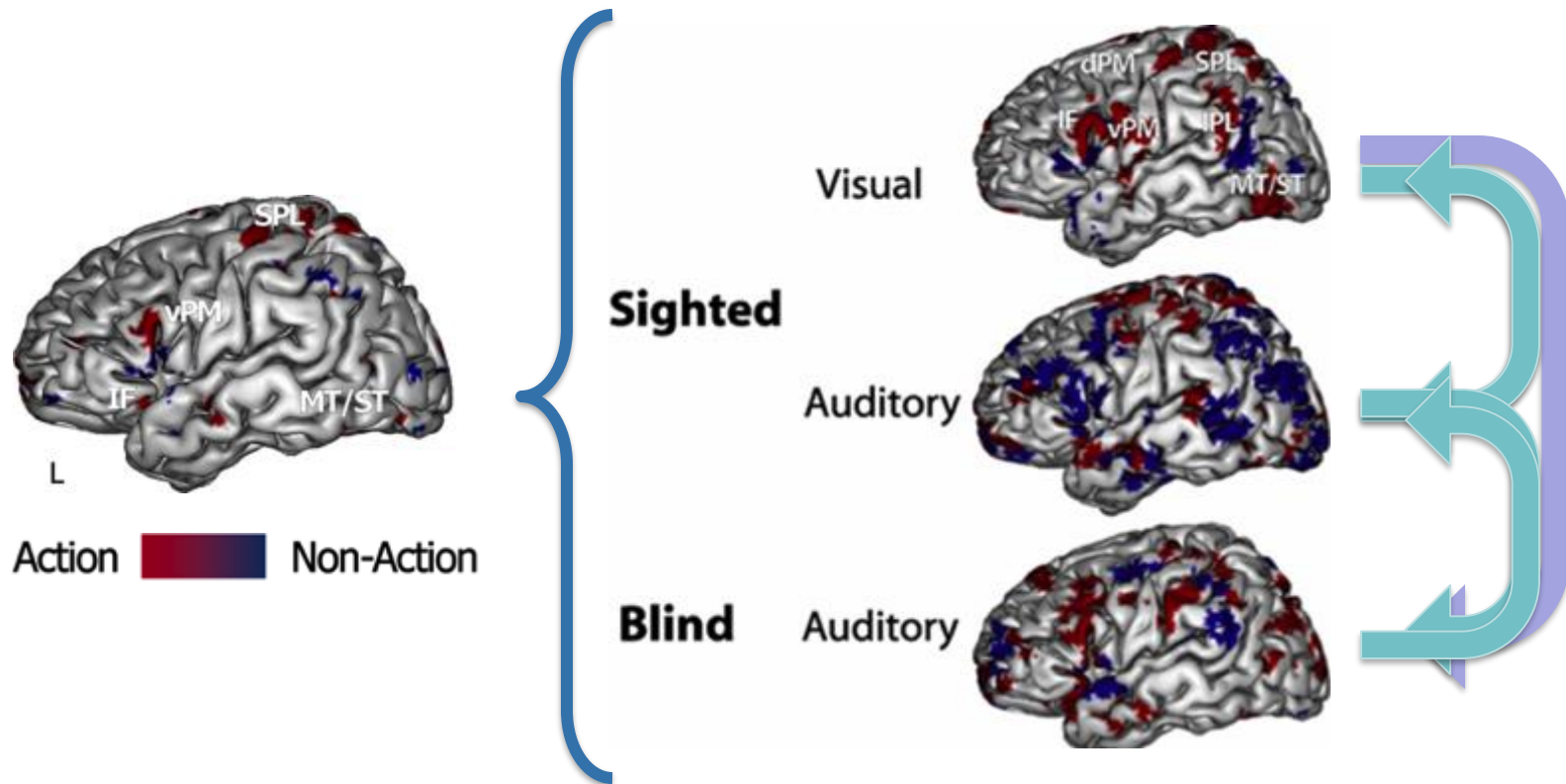
**p-value < 0.01,

*p-value < 0.05 at permutation test.

Beyond motor scheme: A supramodal distributed representation in the action-observation network



Beyond motor scheme: A supramodal distributed representation in the action-observation network



- The representation of the same stimulus category (action) is accurately recognized within and across different sensory modalities (visual, auditory or motor) and experimental groups (sighted and congenitally blind): a truly supramodal functional representation of actions

Beyond Motor Scheme: A Supramodal Distributed Representation in the Action-Observation Network

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		Sighted		Blind
		SVM classifier trained on visual stimuli	SVM classifier trained on auditory stimuli	SVM classifier trained on auditory stimuli
A. Whole brain				
Sighted	Visual	80.7%***	n.s.	n.s.
	Auditory	n.s.	75.7%***	n.s.
Blind	Auditory	n.s.	n.s.	76.2%***
B. After excluding the knock-out map				
Sighted	Visual	77.1%***	n.s.	n.s.
	Auditory	n.s.	74.3%***	n.s.
Blind	Auditory	n.s.	n.s.	73.7%***
C. By restricting to the knock-out supramodal map				
Sighted	Visual	73.6%***	61.1%**	n.s.
	Auditory	59.6%*	67.1%***	57.7%*
Blind	Auditory	60.3%**	61.5%**	70.0%***

***p-value < 0.005,

**p-value < 0.01,

*p-value < 0.05 at permutation test.

How concepts are encoded in the brain



Mammals

Birds

Fruit

Vegetables

Tools

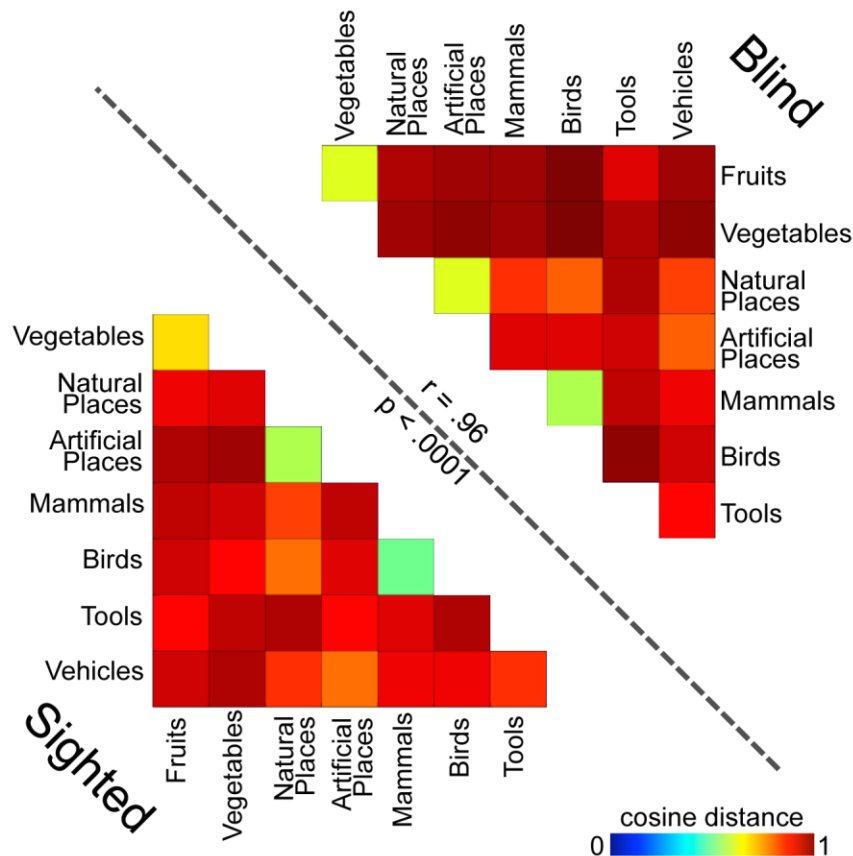
Vehicles

Natural places

Artificial places

Behavioral results

- Representational spaces from behavioral data were correlated across groups



- The forty stimuli were organized in eight semantic categories (vegetables, fruits, natural and artificial places, mammals, birds, tools, vehicles)
- The conceptual representation in sighted and blind is highly similar (correlation 0.96) suggesting **a common linguistic (and thus semantic) representation of distinct object categories**